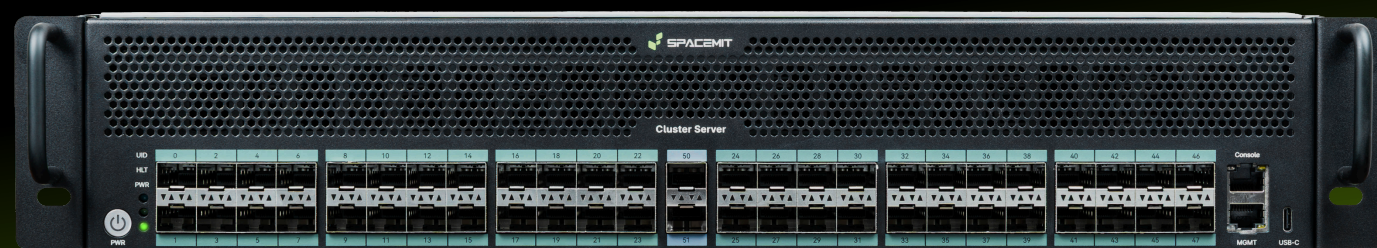


Cluster Server RV2768

RISC-V Architecture For Intelligent Future



768-Core Native RISC-V AI Compute Cluster

The Cluster Server RV2768 is a RISC-V-based cluster server in a 2U, 19-inch rack form factor, integrating up to 48 SpacemiT K3 processors and compliant with the RVA23 profile defined by RISC-V International.

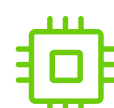
RVA23 targets general-purpose application processors, requiring support for Vector and Hypervisor extensions, and represents a modern performance standard for AI and high-performance workloads. The system delivers 768 native RISC-V cores, making it well-suited for running large numbers of AI agents and applications while handling compute-intensive data processing tasks.

With a multi-node cluster architecture and full hardware virtualization, the server allows multiple users to run independent computing environments simultaneously without interference, providing hardware-level isolation between users. Leveraging vector extensions, it delivers up to 2880 TOPS of AI compute and supports up to 1536 GB of memory, enabling concurrent AI inference workloads.



Flexible Compute Scaling

- 2U chassis supporting up to 48 compute nodes (384 CPU cores + 384 AI cores)
- External switch connectivity to any node via independent, non-blocking channels



Standards-Compliant Architecture

- RVA23 profile support, meeting instruction set requirements for high-end Linux distributions over the next five years and beyond
- Full hardware virtualization across CPU, memory, interrupts, and I/O
- 1024-bit wide parallel AI compute for efficient integration with mainstream AI ecosystems



Reliable & Serviceable

- Independent maintenance, remote access, debugging, and upgrade per compute node
- Front I/O design for simplified servicing and improved thermal efficiency (no rear cabling)
- Rear “N+1” redundant fan design



Intelligent Management

- PXE boot support for diskless startup via network
- Integrated Layer 3 management switch supporting VLAN segmentation, dynamic routing, and link aggregation
- Built-in BMC for system monitoring, thermal control, and compute module management



Specifications

Physical	Form Factor	2U rackmount
	Dimensions	87.4mm× 447.6mm × 816.8mm
	Compute Modules	Up to 24 modules, each with 2 K3 nodes, independently serviceable
	Compute Nodes	Up to 48 physically isolated nodes, each independently manageable and maintainable
Compute Node	CPU Cores	8× X100 cores per K3, SPECint2006 > 9.0/GHz, up to 2.4 GHz
	AI Cores	8× A100 cores per node, 60 TOPS @ INT4
	RAM	LPDDR5, 8 / 16 / 32 GB options, up to 6400 MT/s
	ROM	NVMe SSD (PCIe 3.0 x4), 64 / 128 / 256 GB, optional UFS
	Networking	1GE port for internal management network, 10GE port for external connectivity
	Features	PXE boot, remote access, debugging, and upgrade
Switch System	External Ports	2 × 10GE SFP28 optical ports and 48 × 10GE SFP+ optical ports
	Topology	Network connectivity between compute node interfaces, sideband management ports, and external ports
System (Chassis)	Power	2000 W / 3000 W PSU (Platinum/Titanium), hot-swappable, 1+1 redundancy
	Cooling	4 fans
	Installation	L-type rail support
Management	System Management	Unified multi-node module management and integrated switch management, with a dedicated front GE management port for full fault monitoring and operations
	Node Management	Power control, reset, firmware configuration, serial debug, and remote console
Software	OS	Optional pre-installed mainstream Linux distributions
	Software Platform	Cluster management, distributed computing, AI agent clusters
Operating Conditions	Temperature	5°C~40°C (41°F~104°F), ASHRAE A1/A2/A3 compliant
	Humidity	8%~90%

